

Redes de Transmisión de Datos

Tema 1

Introducción

Objetivos de un Sistema Teleinformático (ST)

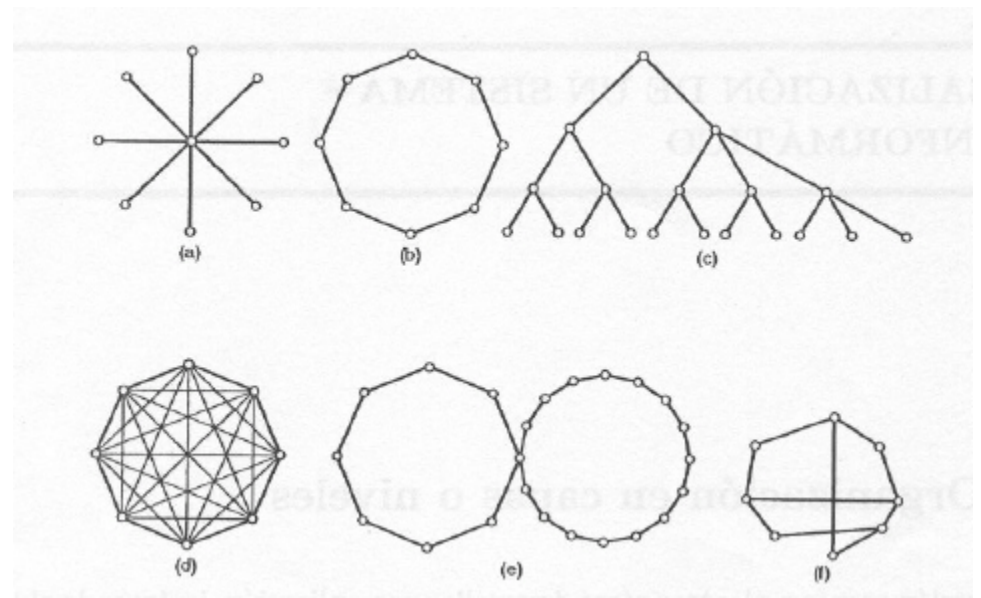
- Compartir los recursos, software o hardware
- Mejorar la fiabilidad
- Abaratar costes

Clasificación de los sistemas teleinformáticos

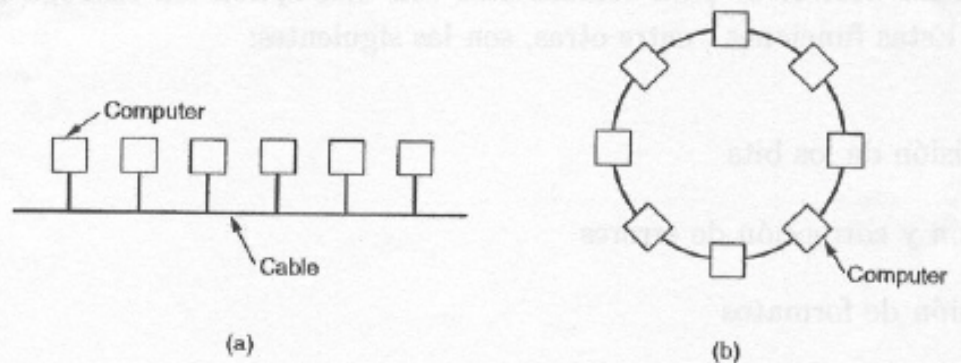
Interprocessor distance	Processors located in same	Example
0.1 m	Circuit board	Data flow machine
1 m	System	Multicomputer
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1,000 km	Continent	
10,000 km	Planet	The Internet

Topologías para la subred

Punto a punto:



Difusión
(*broadcasting*)



Organización en niveles

Filósofo
Marciano



Filósofo
Capuchino
Descalzo



Organización en niveles

Filósofo
Marciano



Comunicación virtual
(protocolo)

← - - - - - →



Filósofo
Capuchino
Descalzo

Organización en niveles

Filósofo
Marciano



Comunicación virtual
(protocolo)
← - - - - - →

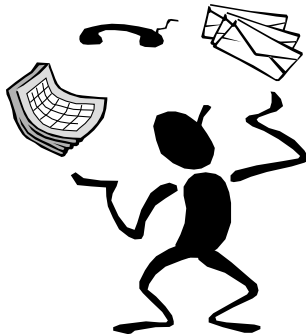
Filósofo
Capuchino
Descalzo



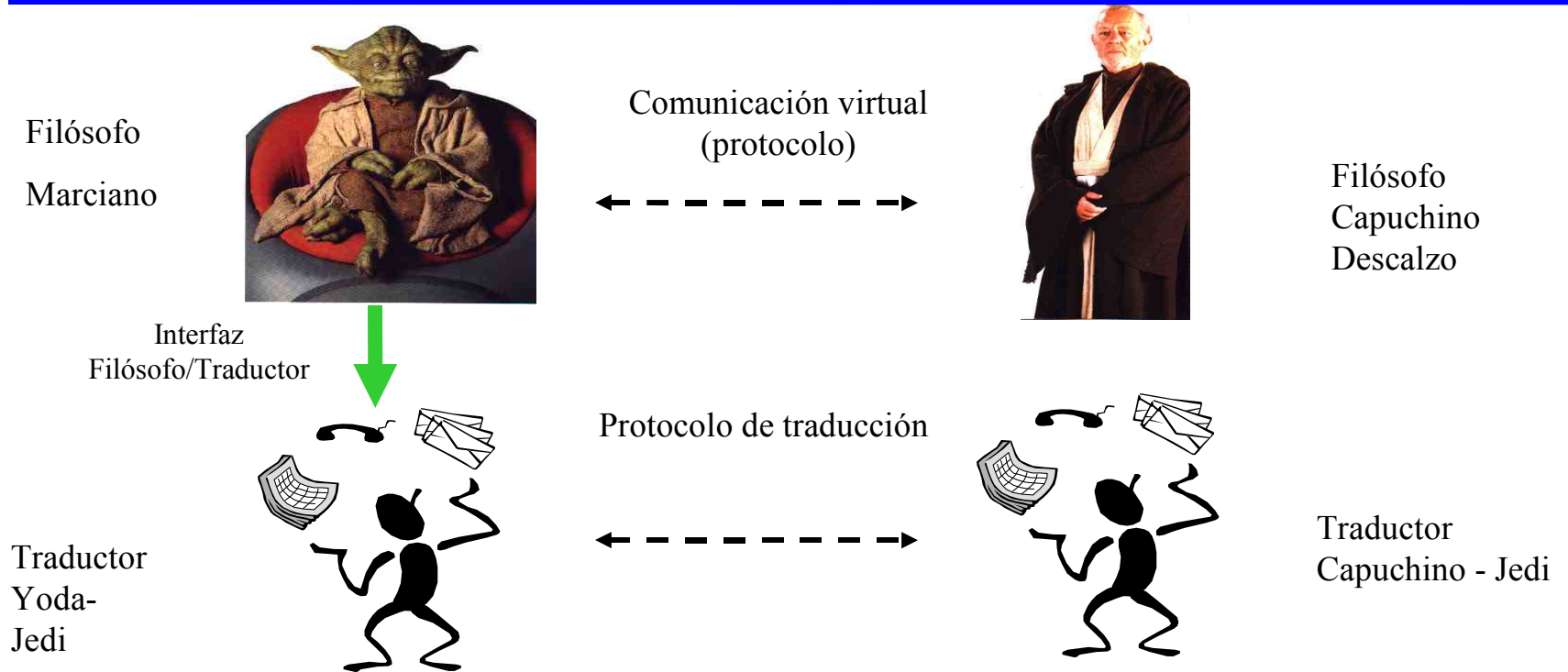
Interfaz
Filósofo/Traductor



Traductor
Yoda-
Jedi



Organización en niveles



Organización en niveles

Filósofo
Marciano



Comunicación virtual
(protocolo)



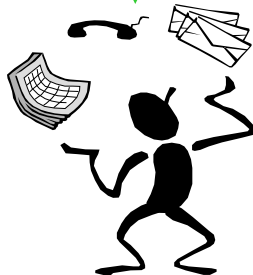
Filósofo
Capuchino
Descalzo



Interfaz
Filósofo/Traductor



Traductor
Yoda-
Jedi



Protocolo de traducción



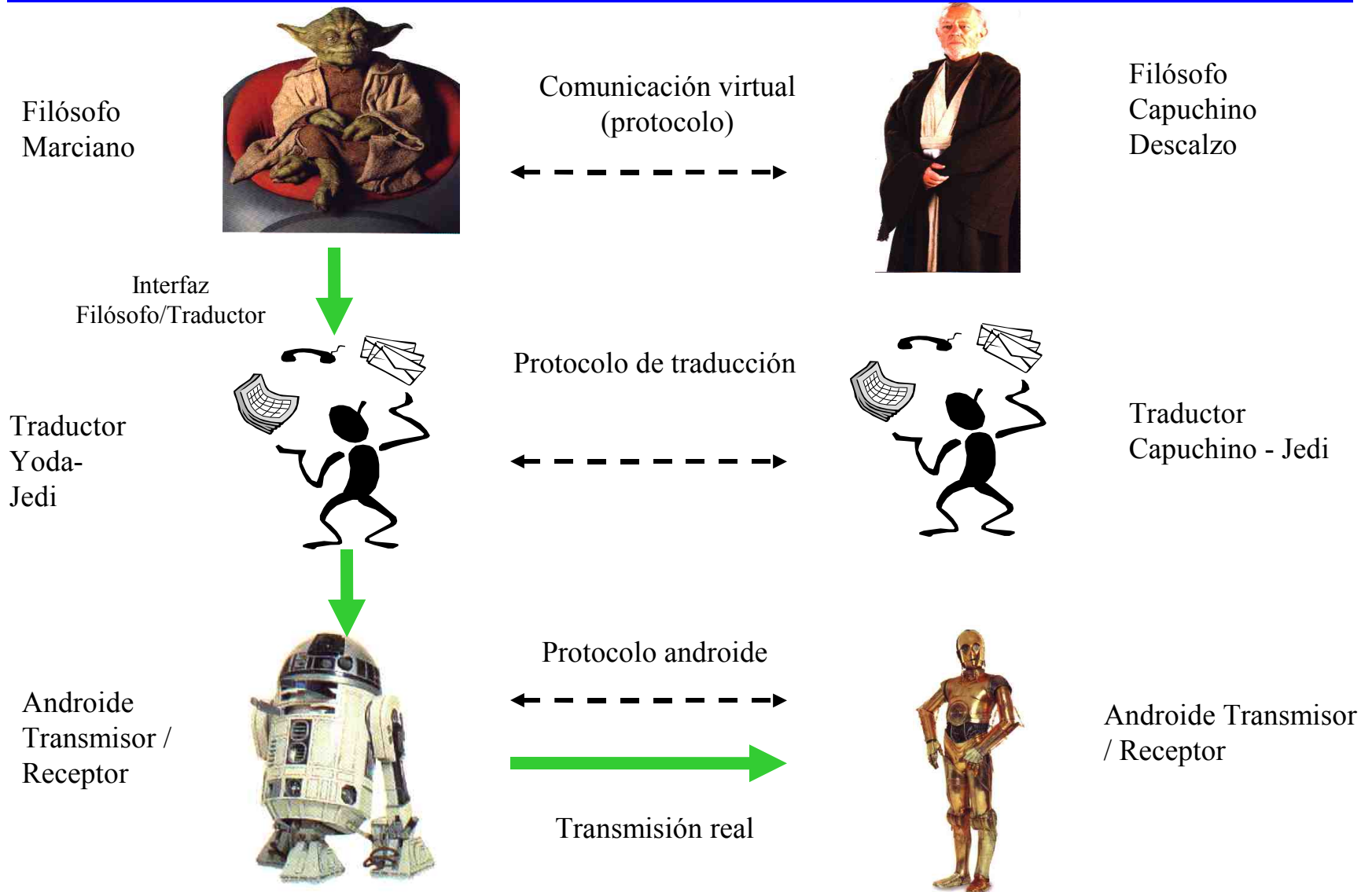
Traductor
Capuchino - Jedi



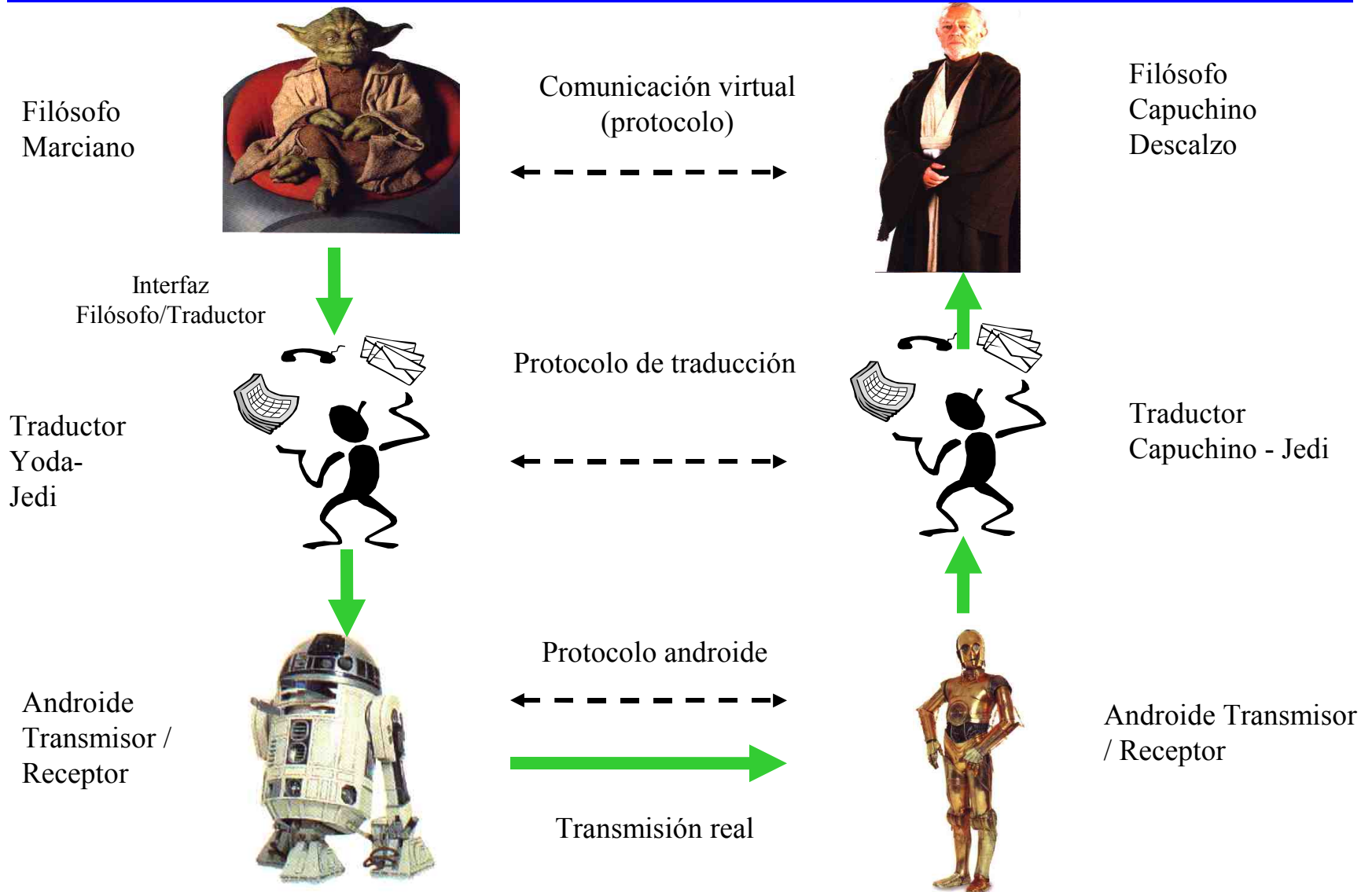
Androide
Transmisor



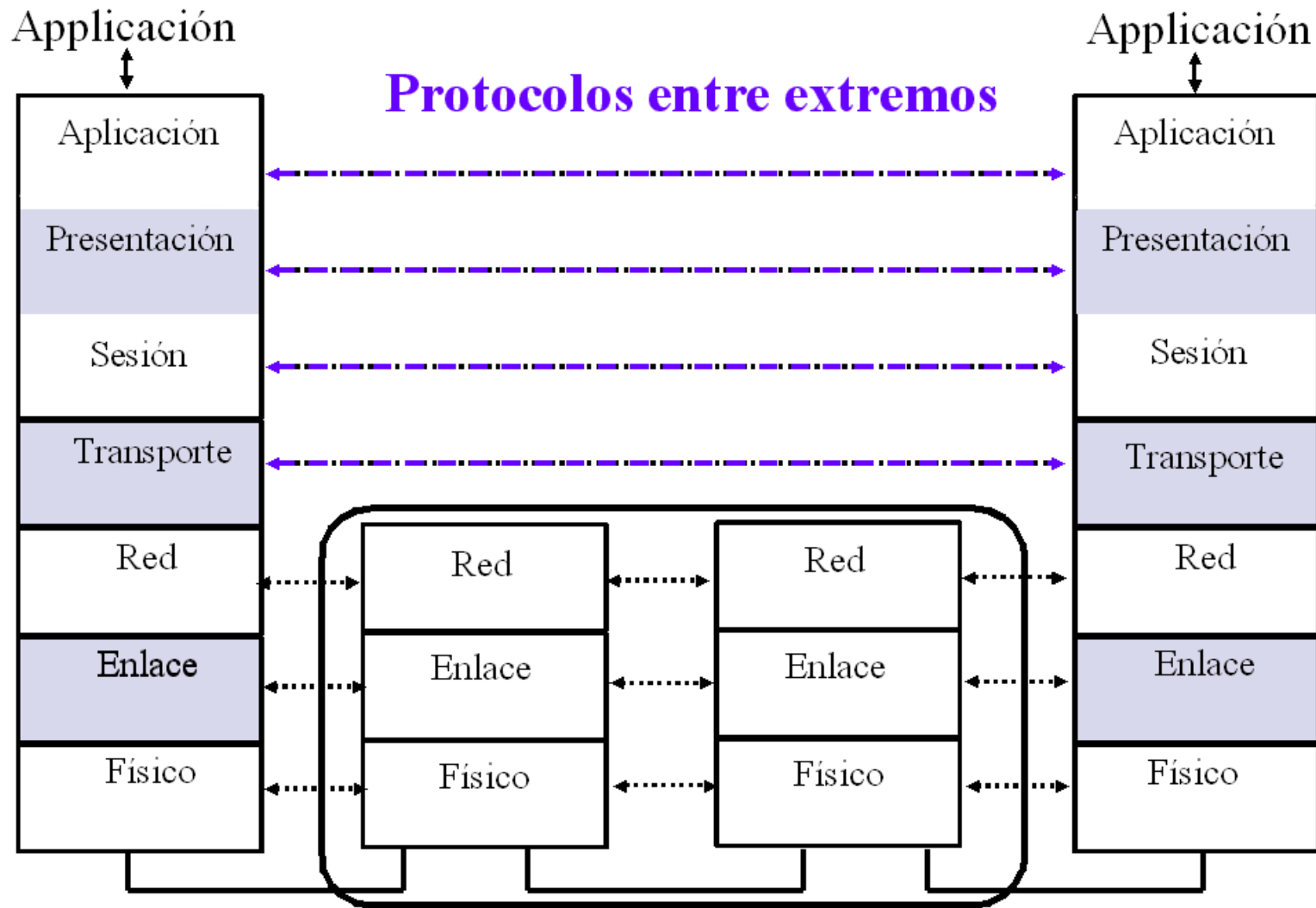
Organización en niveles



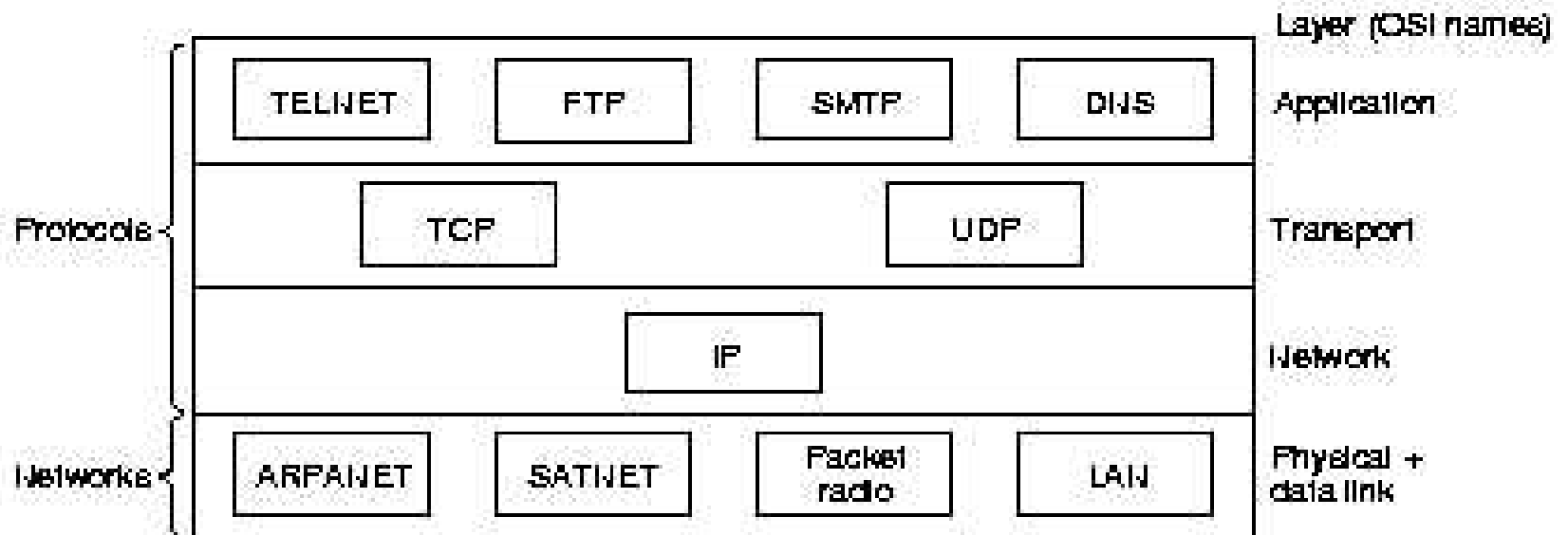
Organización en niveles



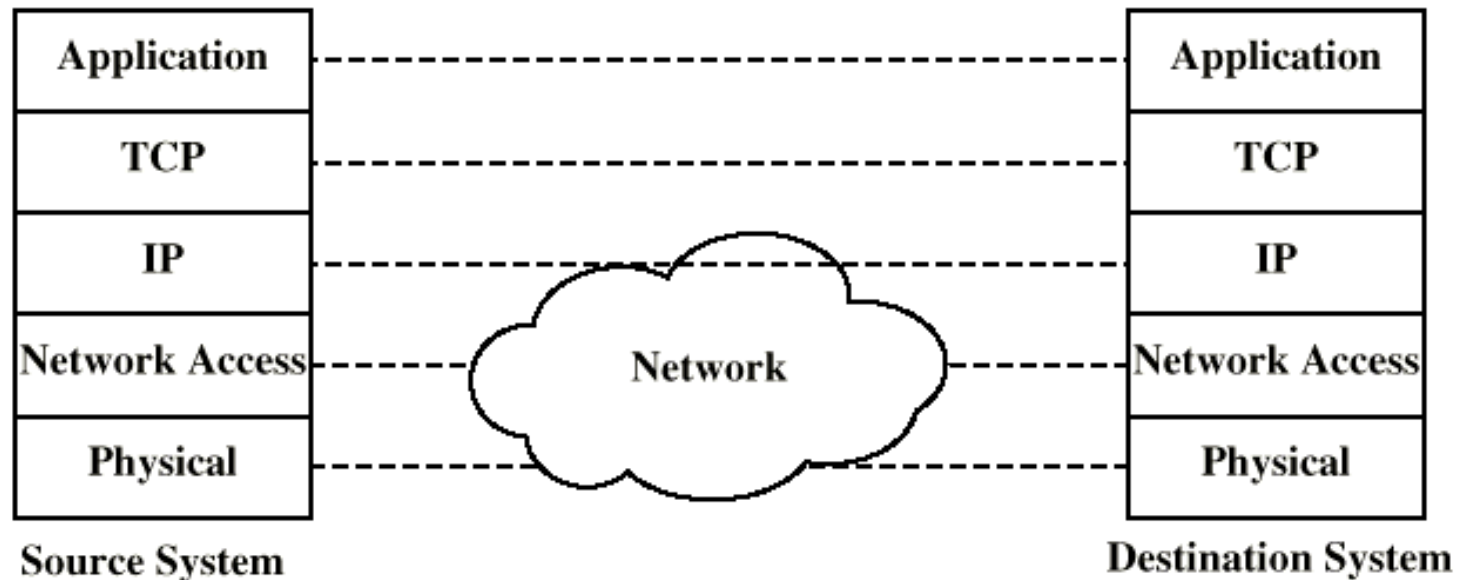
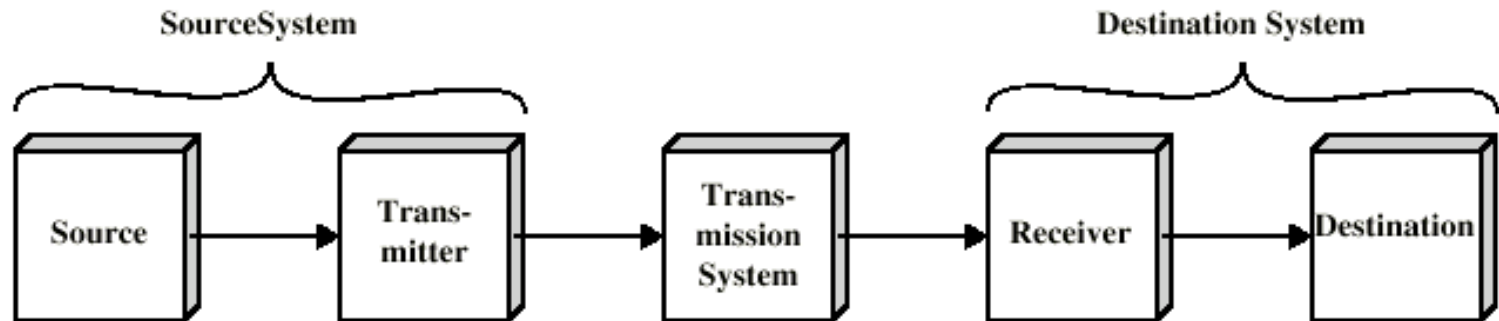
Modelo ISO/OSI



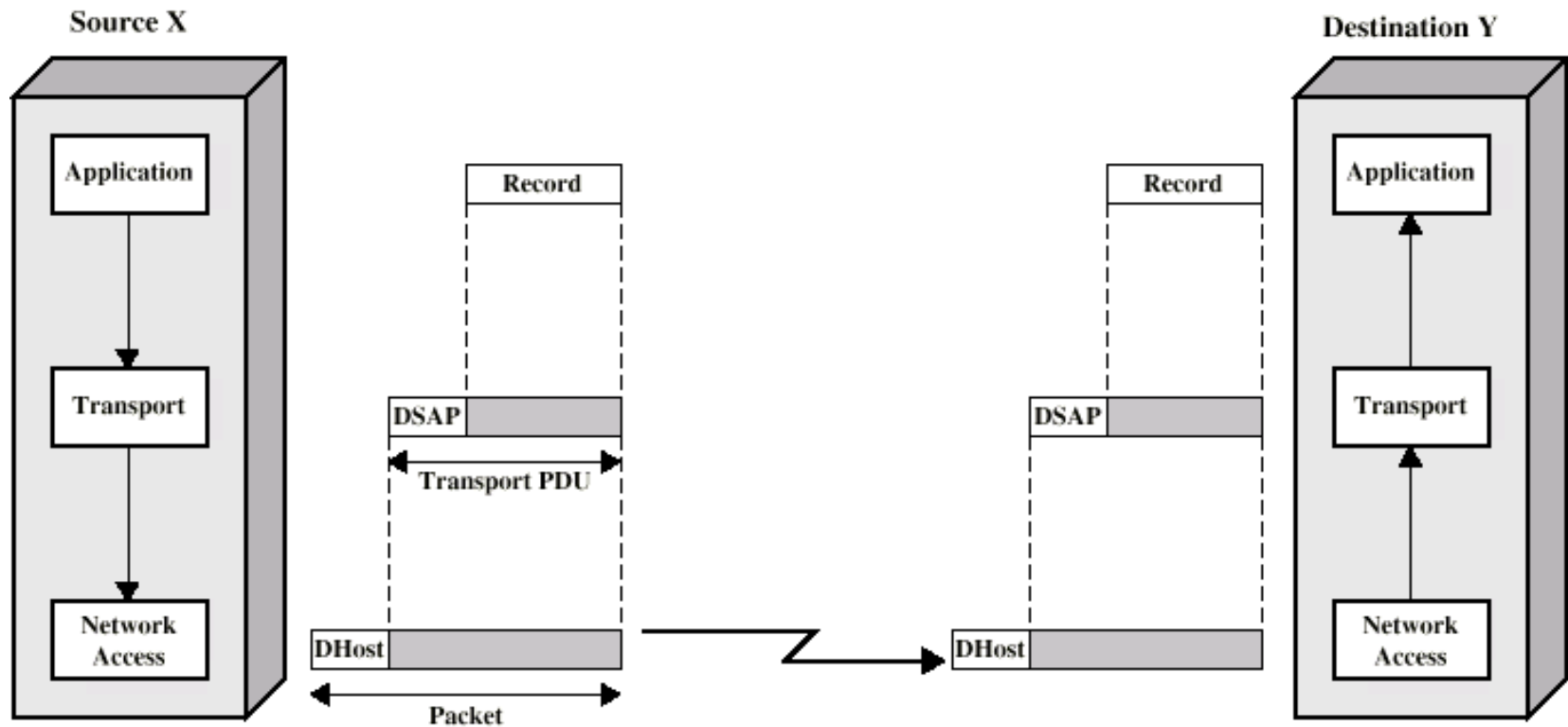
MODELO TCP/IP



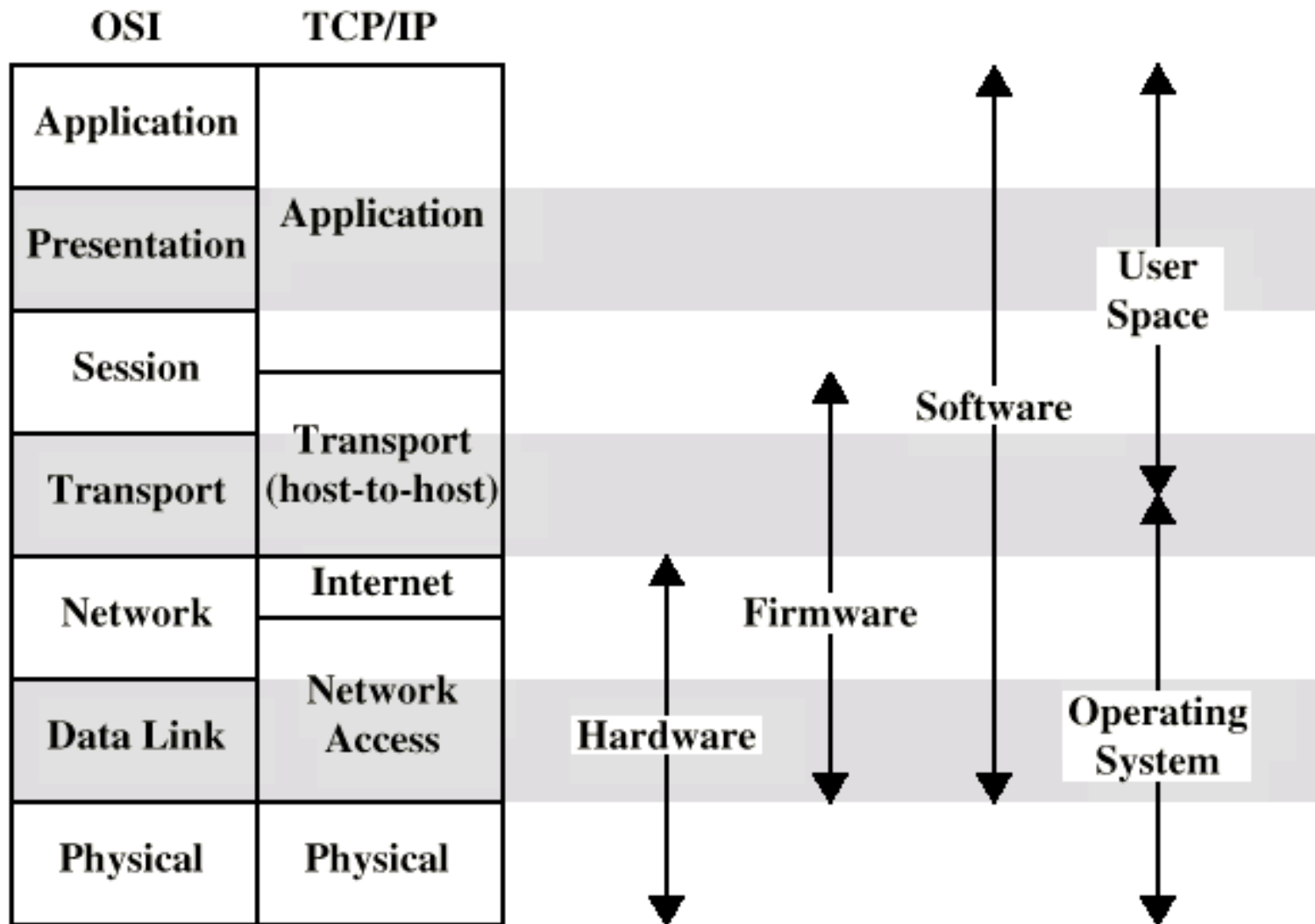
MODELO TCP/IP



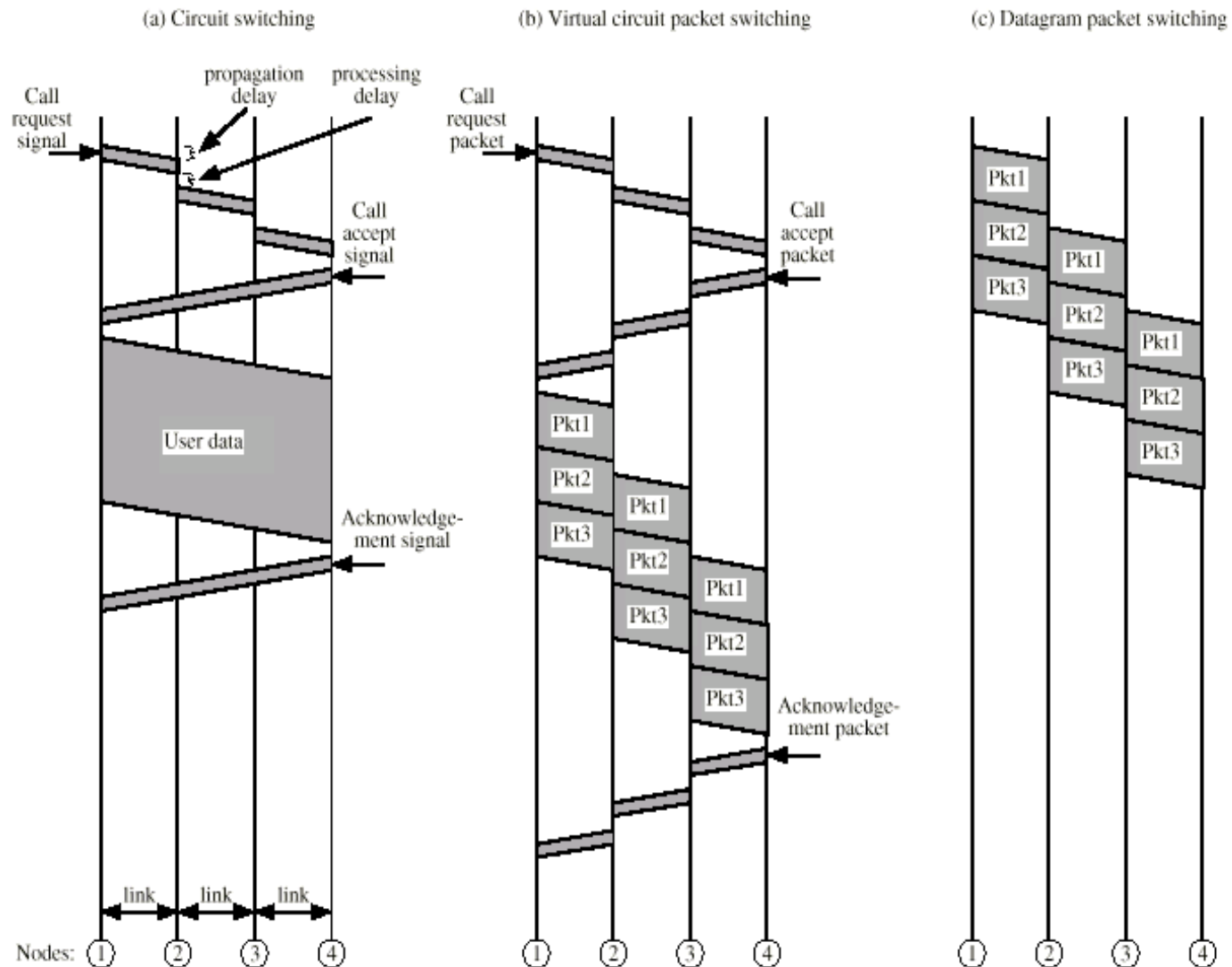
Operation of a Protocol Architecture



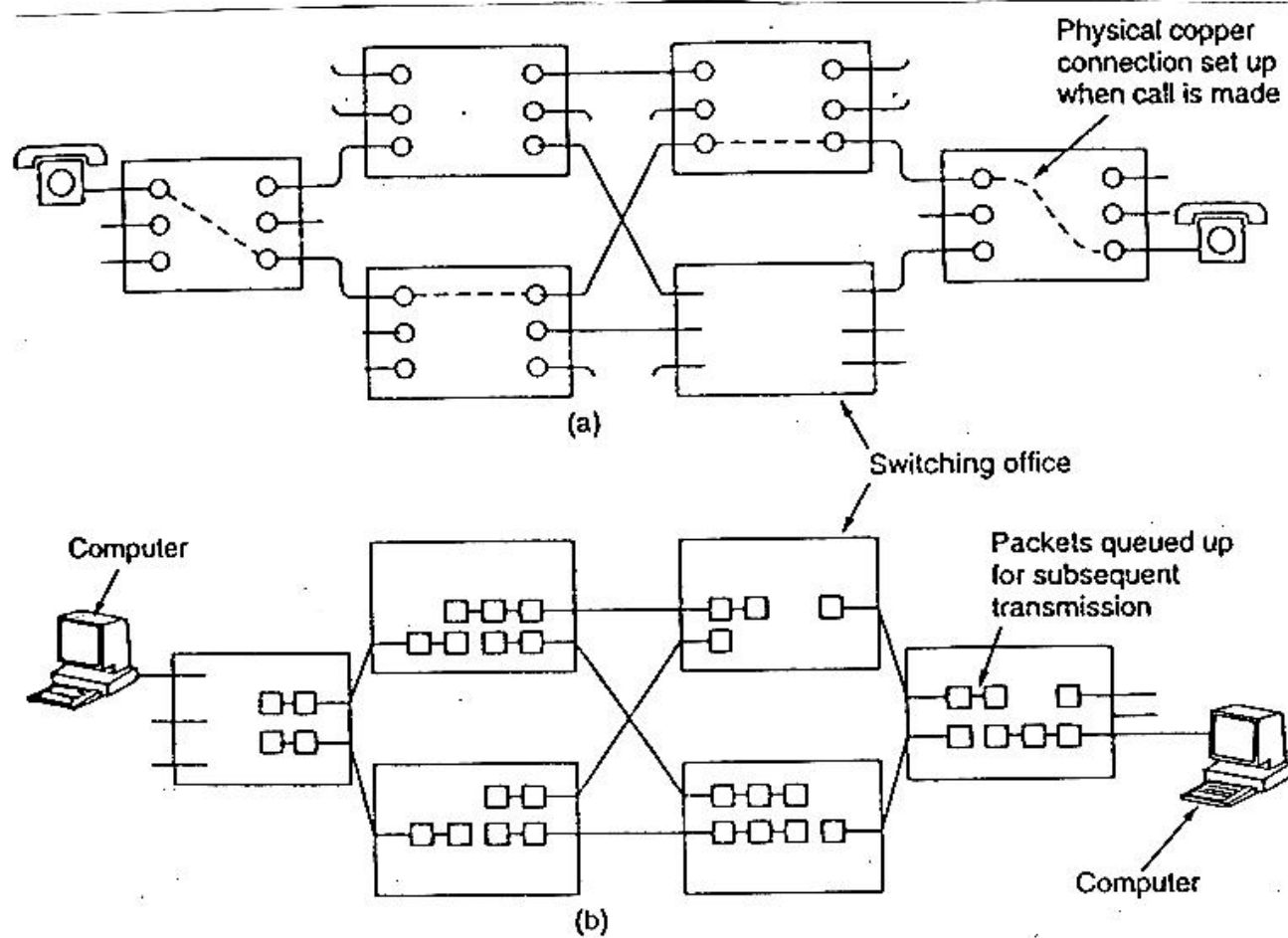
COMPARACIÓN OSI - TCP/IP



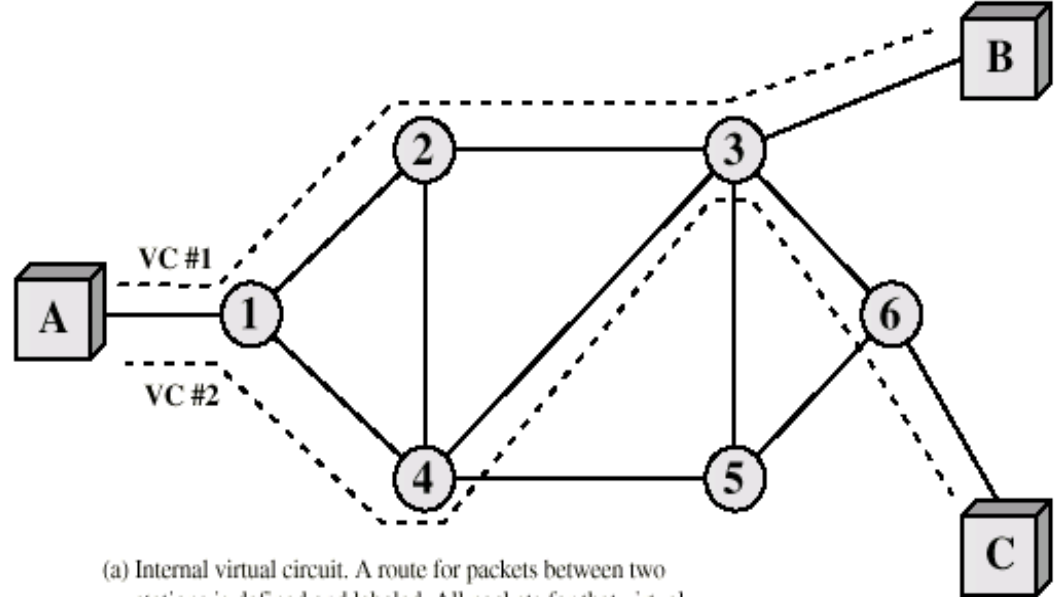
Tipos de conmutación



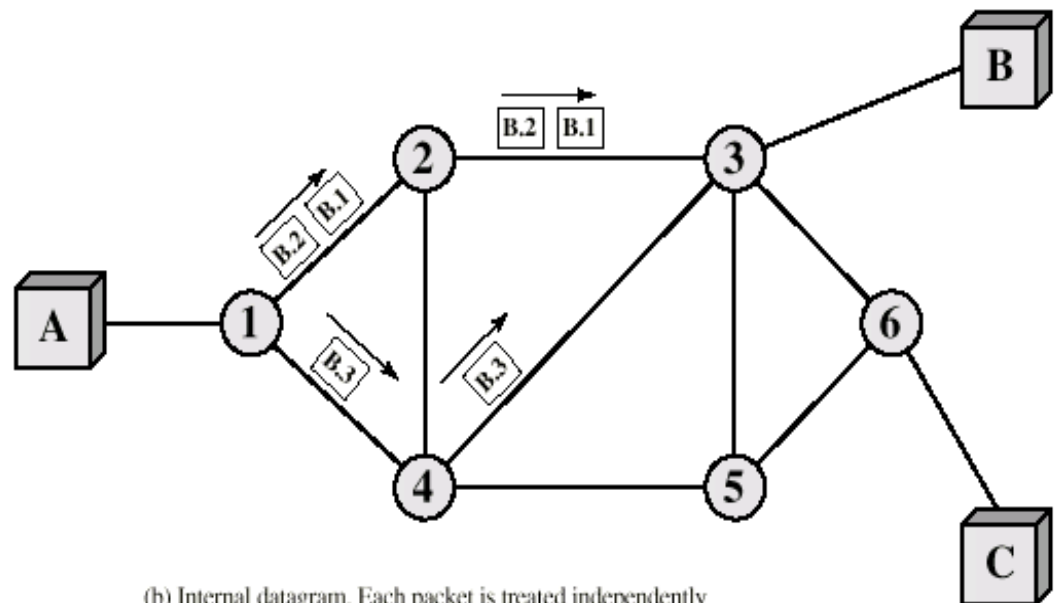
Transmisión según conmut.



Circuitos virtuales

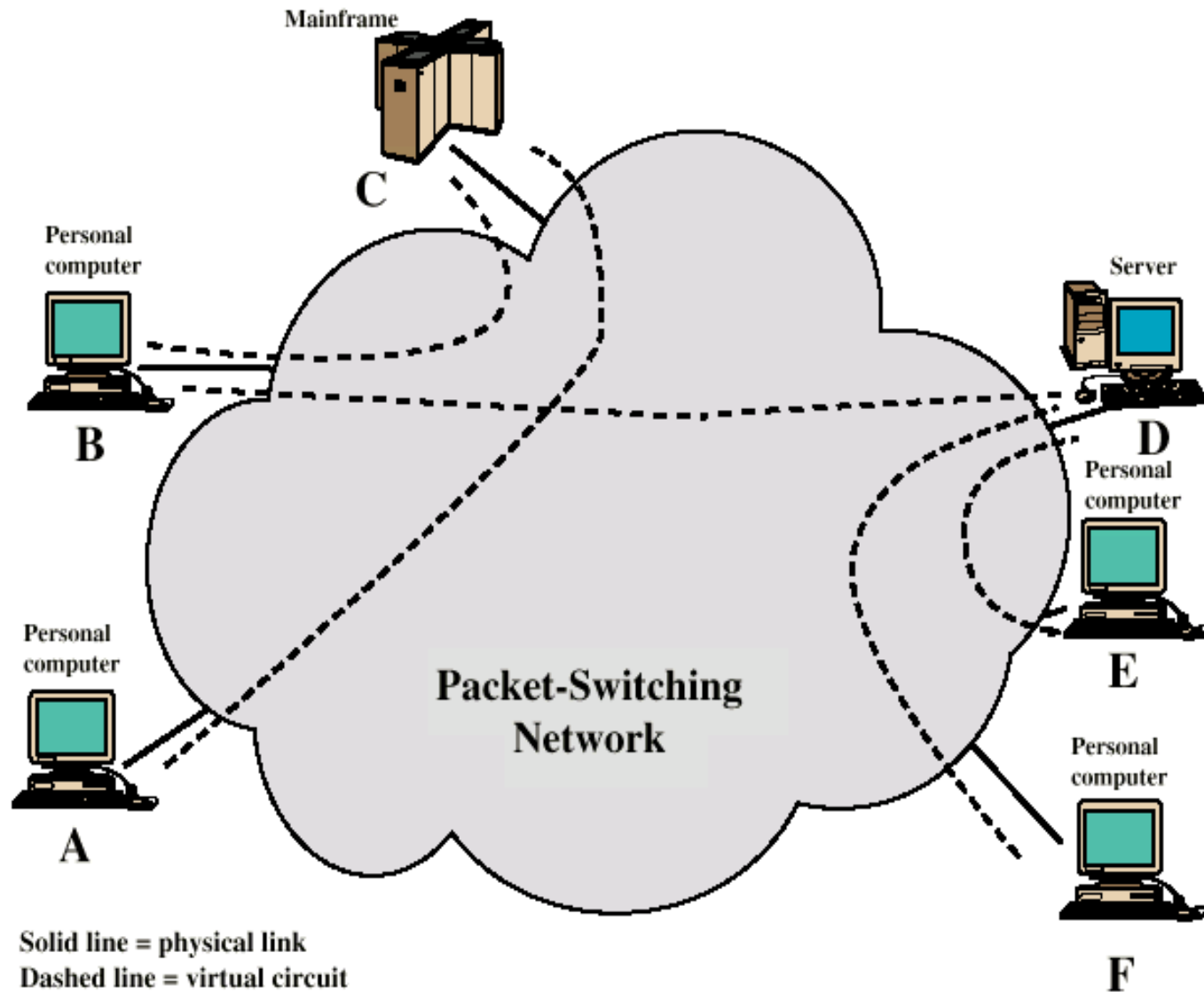


(a) Internal virtual circuit. A route for packets between two stations is defined and labeled. All packets for that virtual circuit follow the same route and arrive in the same sequence.

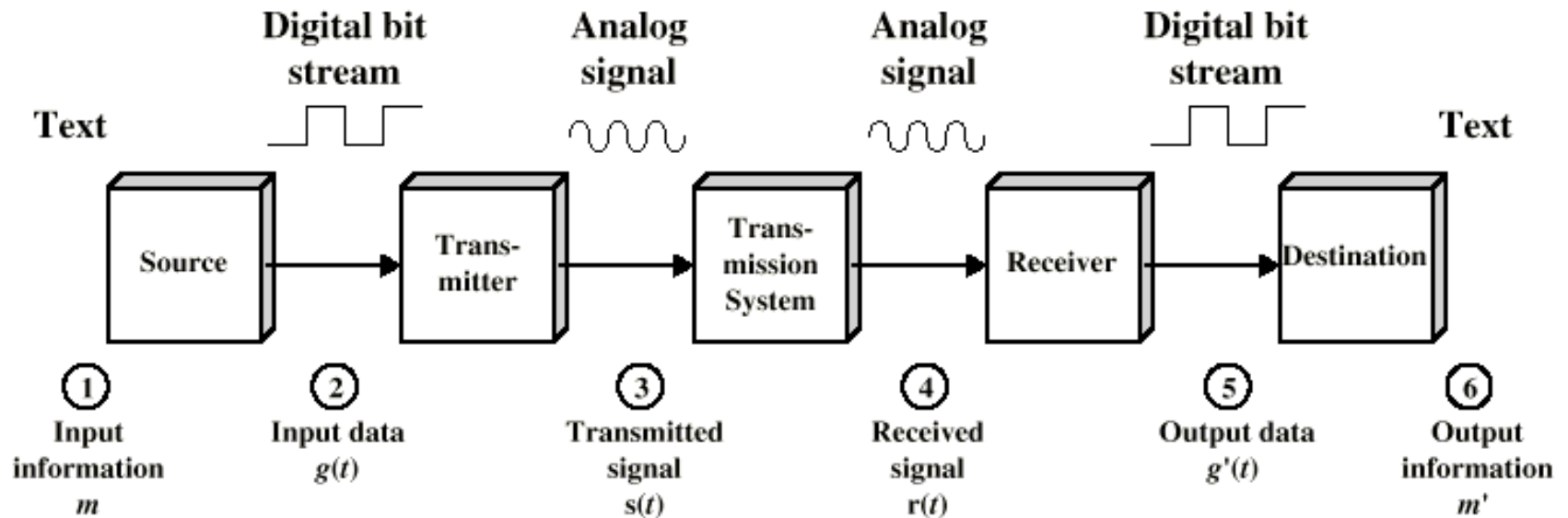


(b) Internal datagram. Each packet is treated independently by the network. Packets are labeled with a destination address and may arrive at the destination node out of sequence.

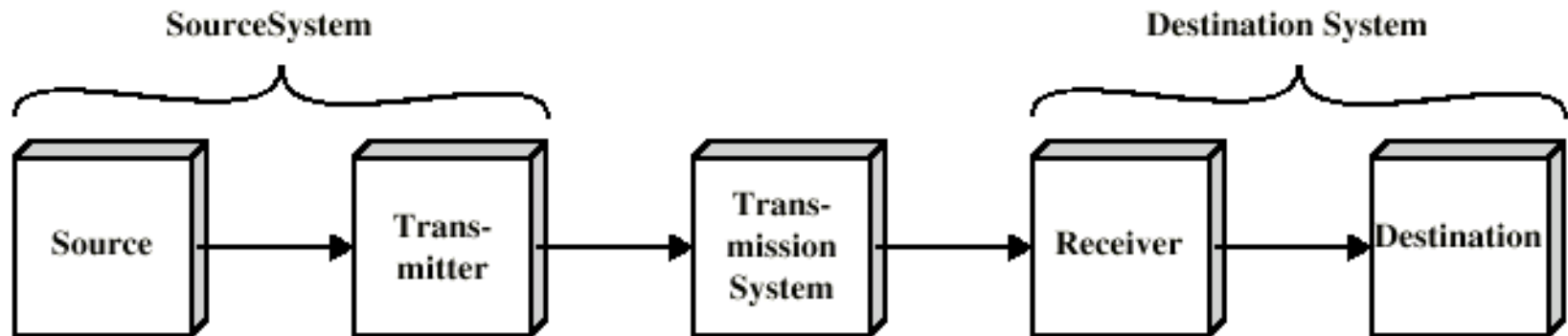
Circuitos virtuales en X.25



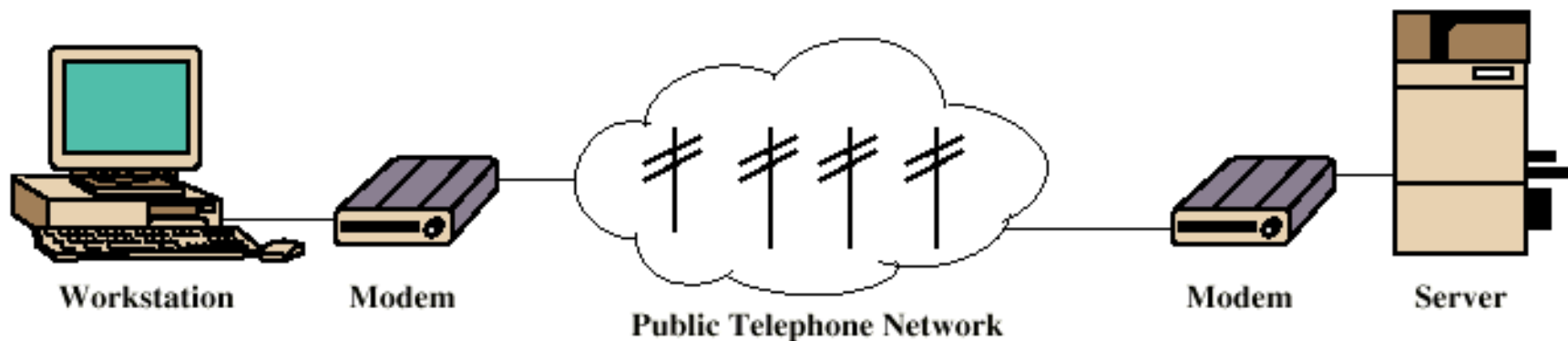
Modelo de un sistema de comunicación (SC)



Modelo de un sistema de comunicación

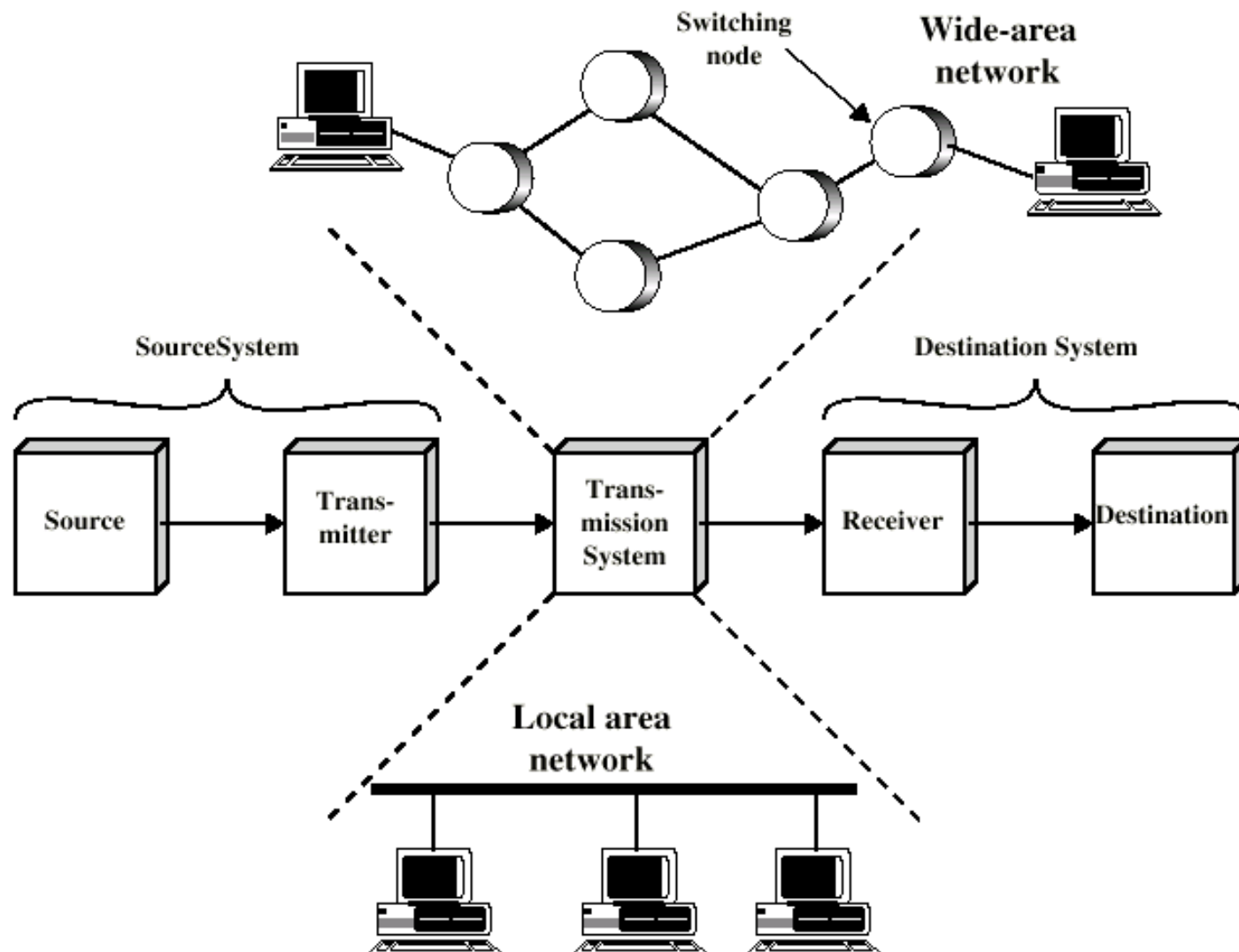


(a) General block diagram



(b) Example

Ejemplo de red de computadores como SC



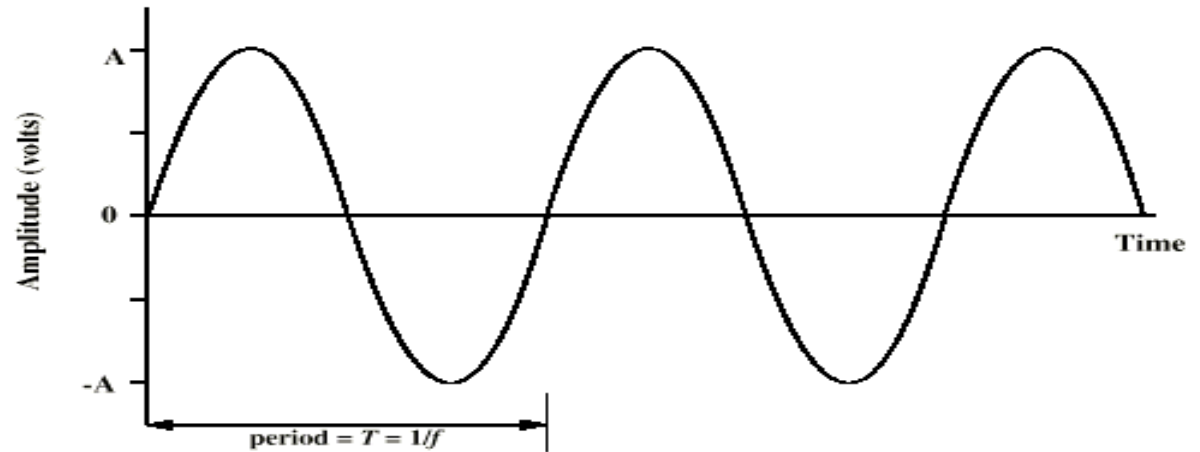
Capa física de la ISO

- Se ocupa del interfaz físico entre el dispositivo transmisor de datos (computador) y el medio de transmisión o la red.
- Define las características del medio de transmisión usado, tales como:
 - Niveles de tensión
 - Tipos de conectores
 - Velocidades de transmisión y longitud del enlace
 - etc.

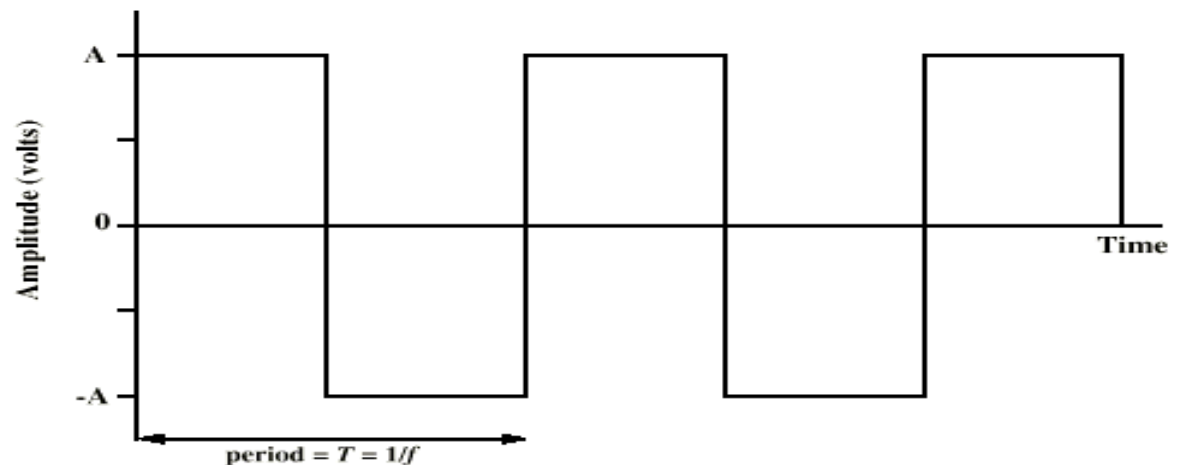
Conceptos básicos sobre análisis de señales

Amplitud (A)

Periodo (T)



(a) Sine wave



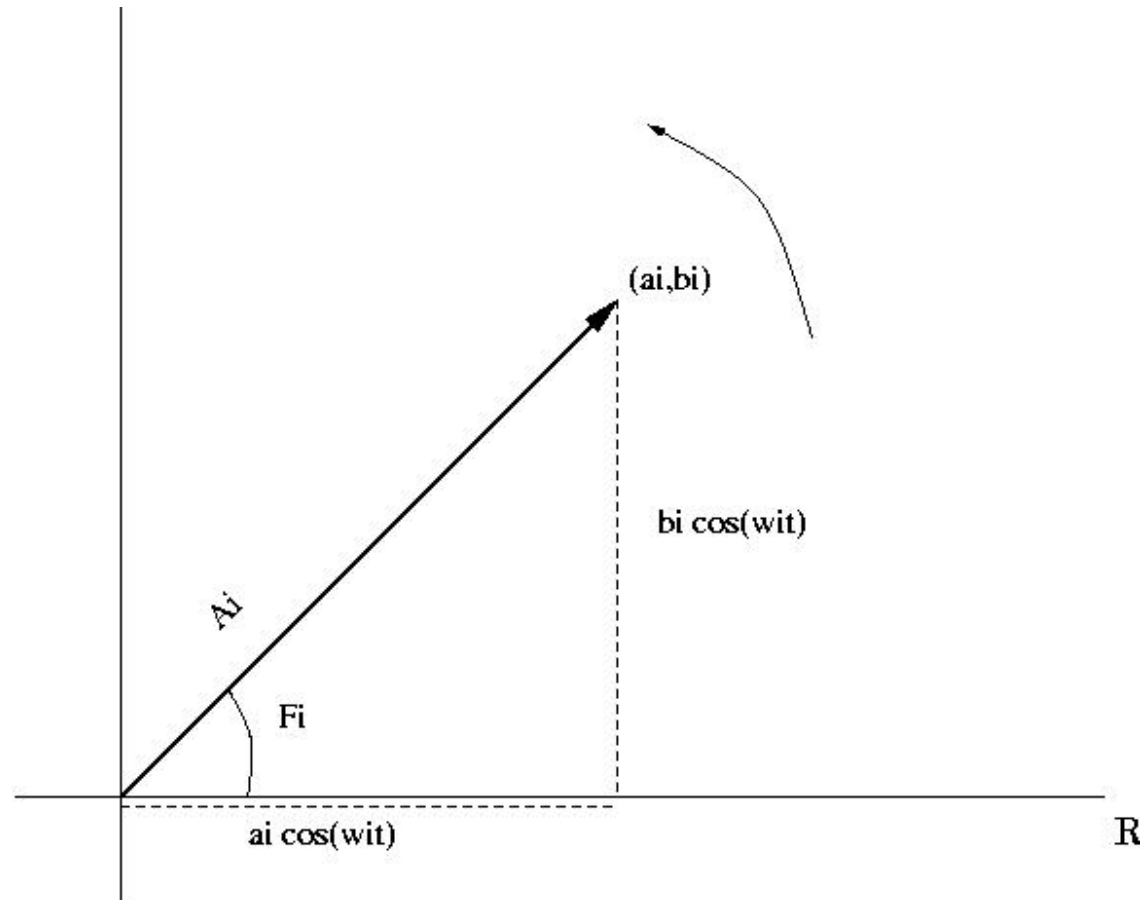
(b) Square wave

Conceptos básicos sobre análisis de señales

Fasor o armónico de orden i

$$a_i \cdot \cos(w_i t) + b_i \cdot \sin(w_i t)$$

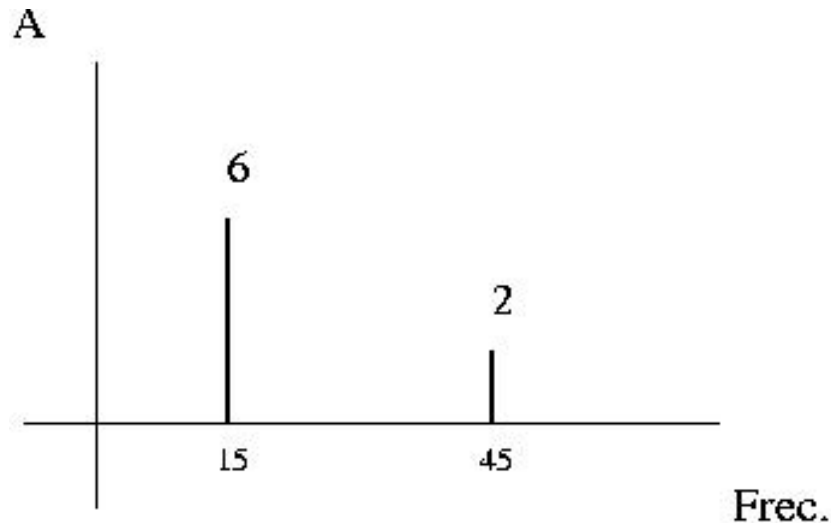
$$A_i = \sqrt{a_i^2 + b_i^2}$$



Conceptos básicos sobre análisis de señales

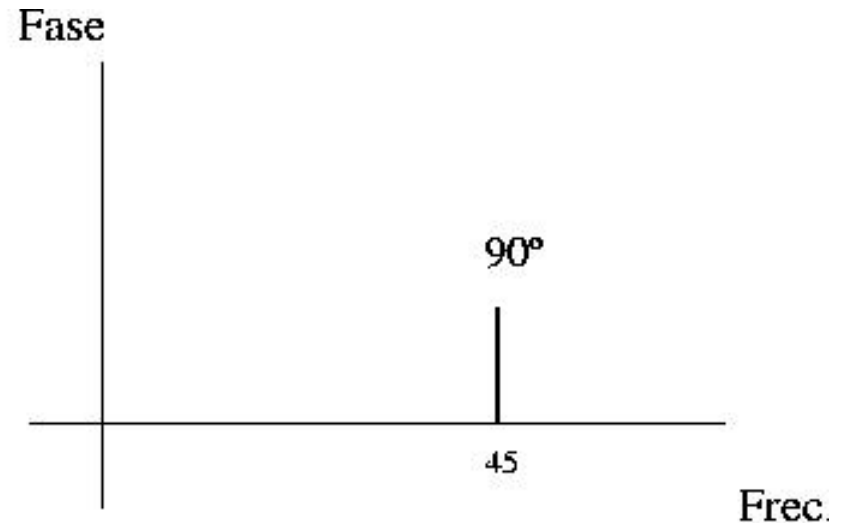
$$x(t) = 6 \cdot \cos(2\pi 15t) + 2 \cdot \cos(2\pi 45t + 90^\circ)$$

Espectro de amplitud (a)

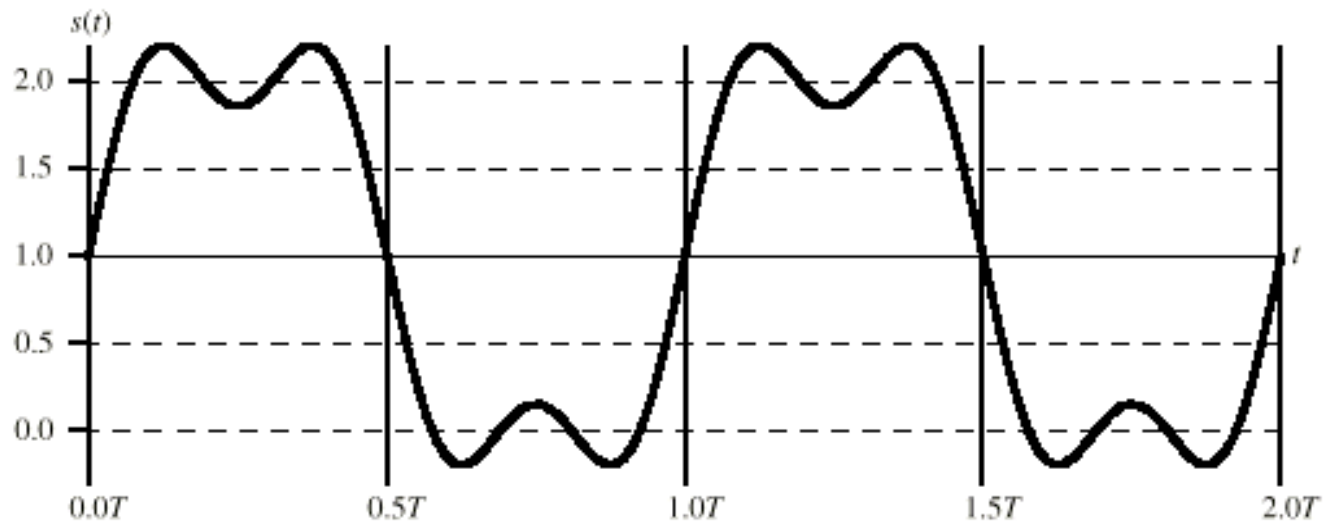


(a)

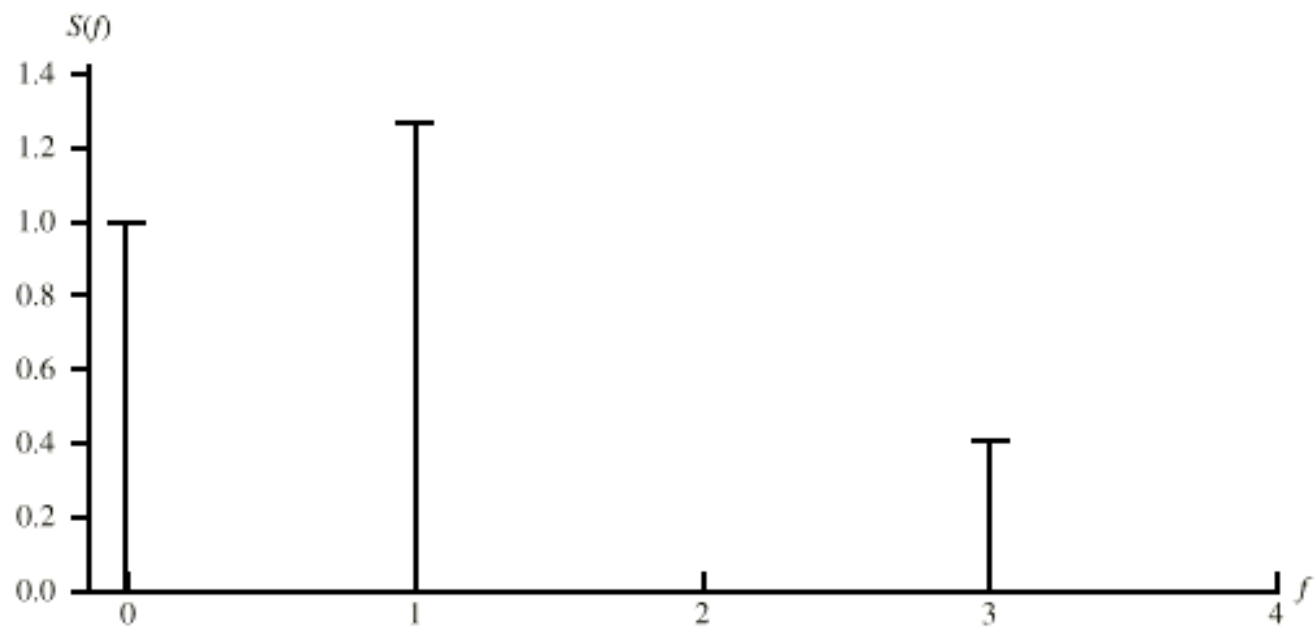
Espectro de Fase (b)



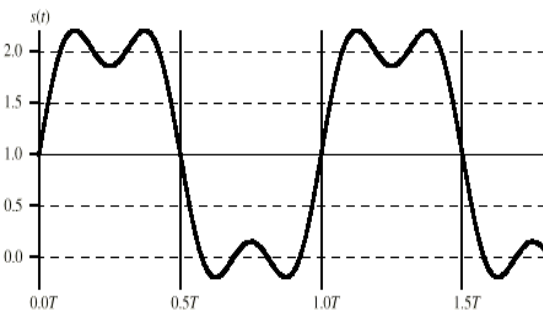
(b)



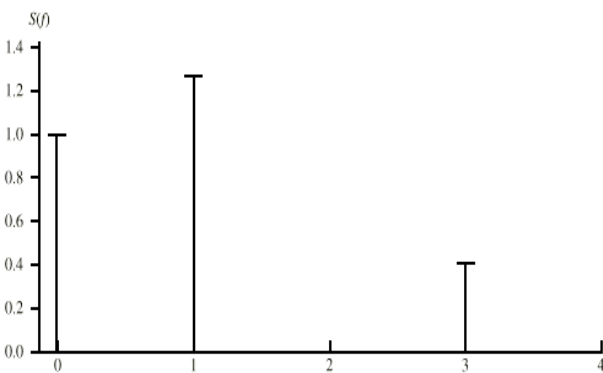
(a) $s(t) = 1 + (4/\pi) [\sin(2\pi ft) + (1/3) \sin(2\pi (3f)t)]$



(b) $S(f)$

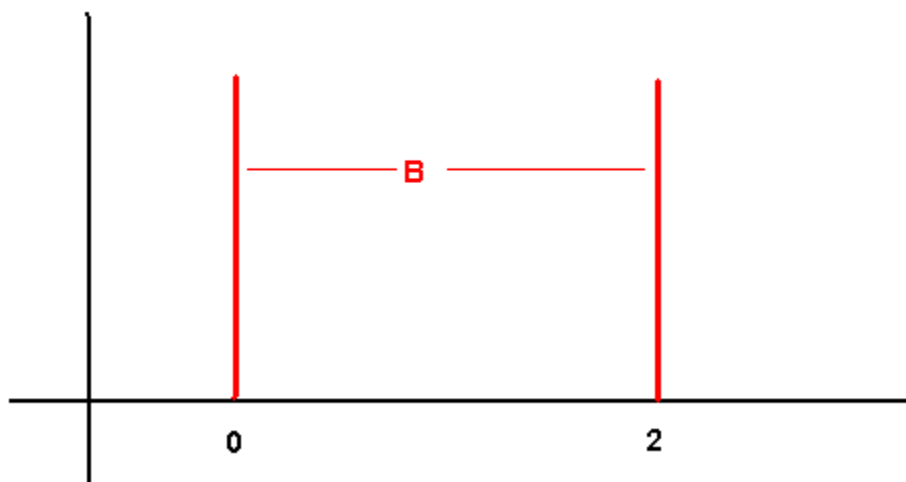


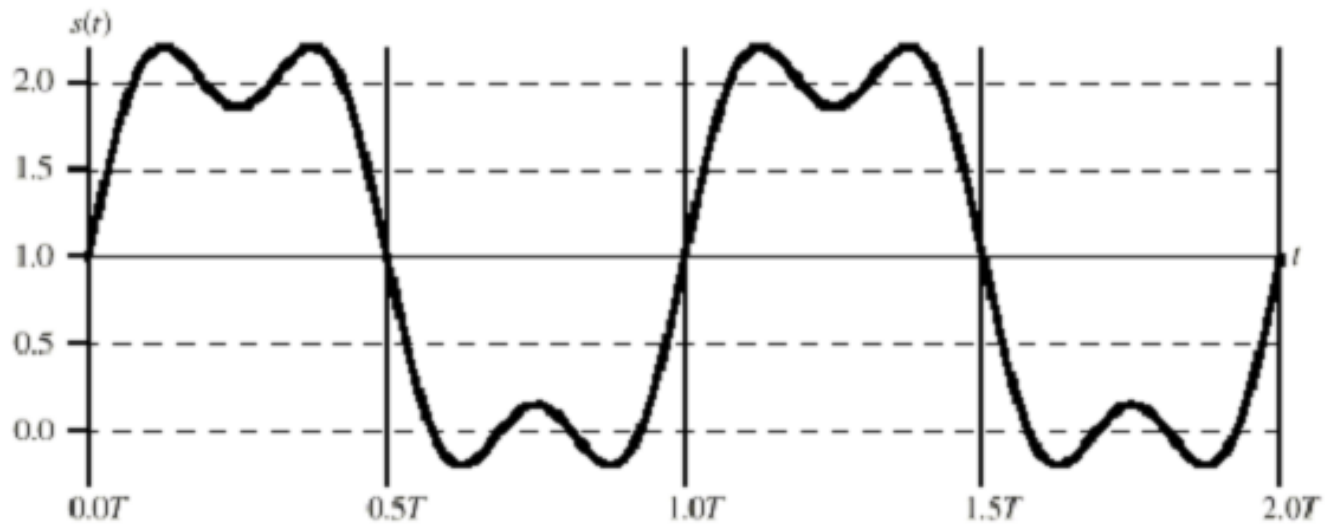
(a) $s(t) = 1 + (4/3) [\sin(2\pi f t) + (1/3) \sin(2\pi 3f t)]$



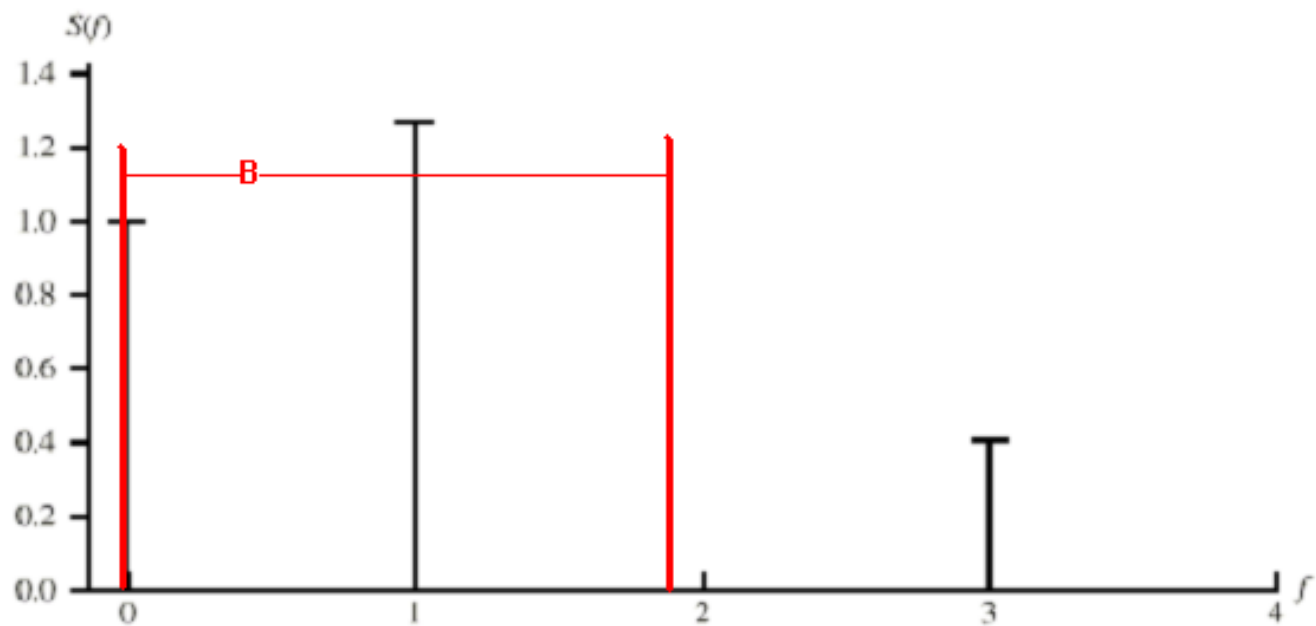
(b) $S(f)$

Extremo Emisor **CANAL TRANSMISIÓN** **Extremo Receptor**

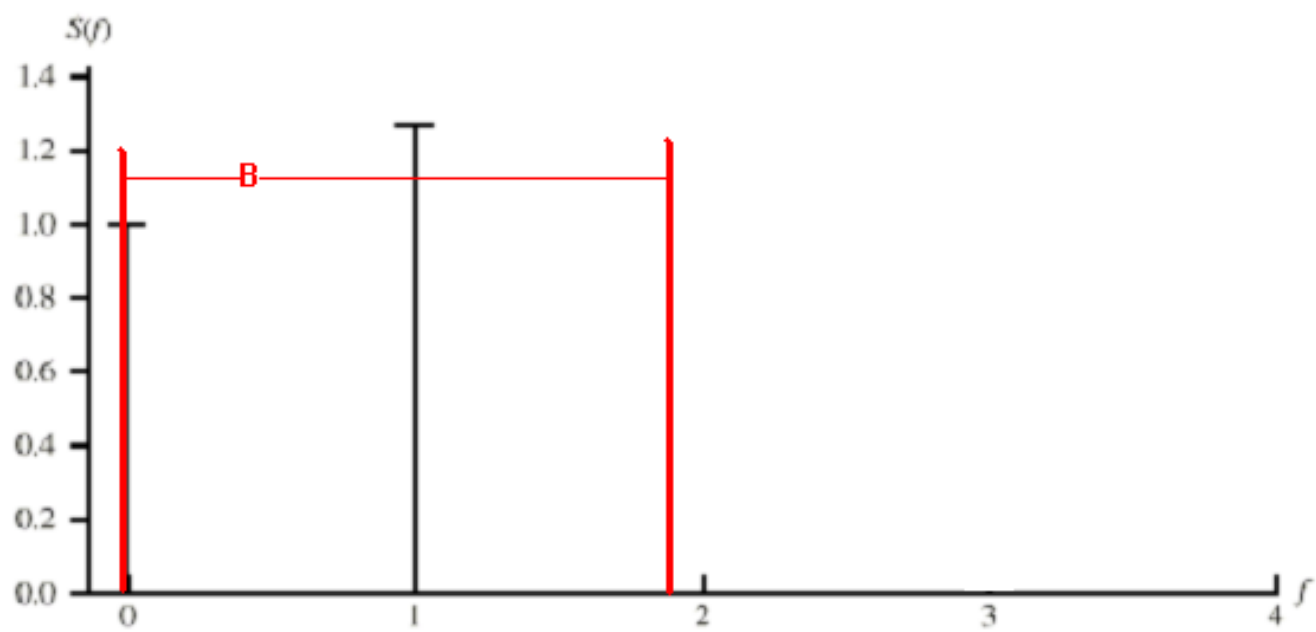




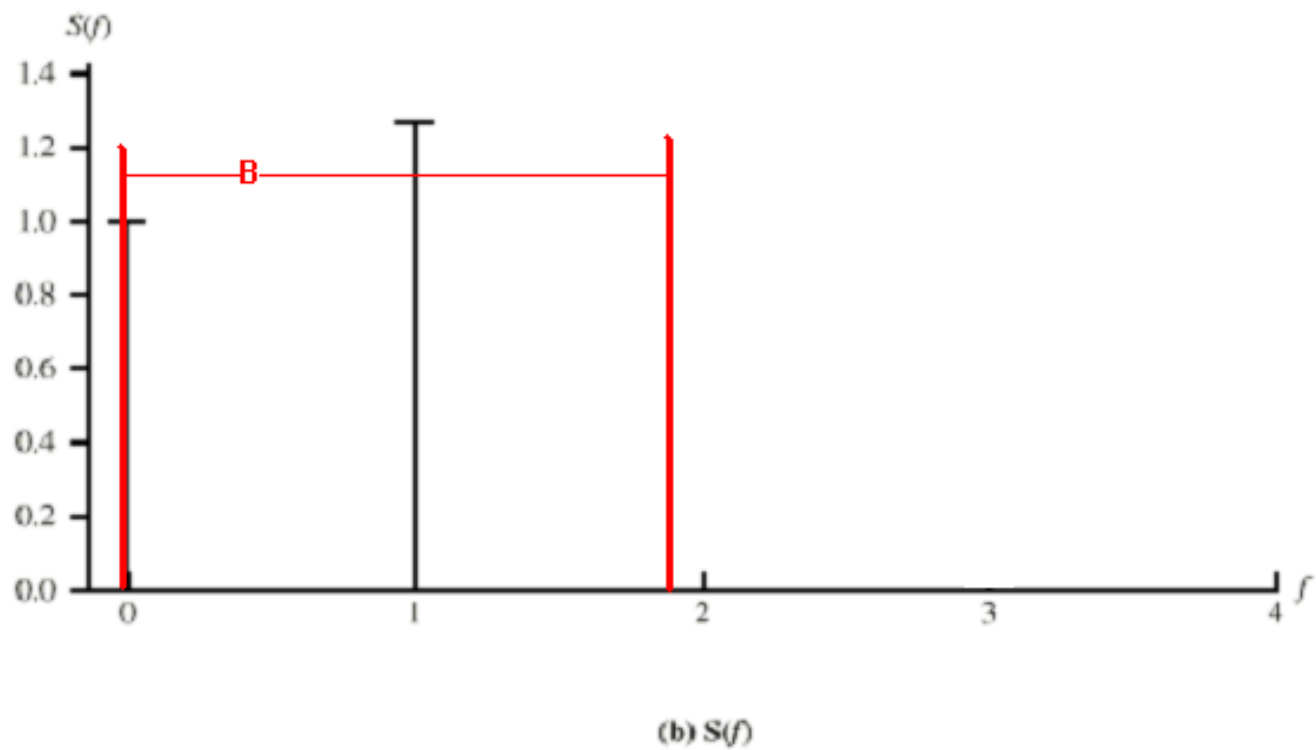
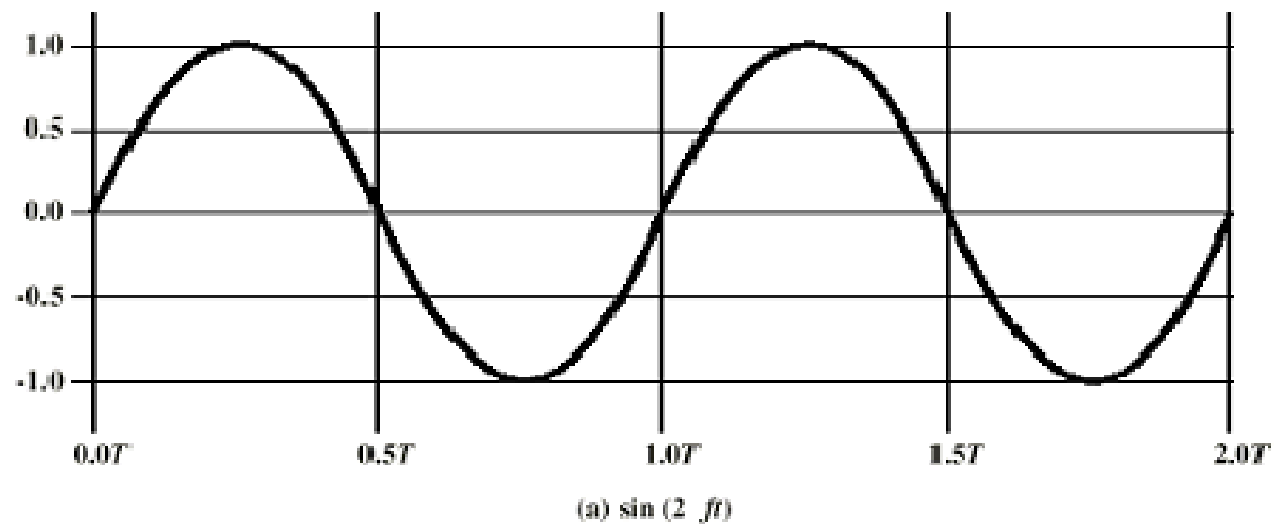
$$(a) s(t) = 1 + (4/\pi) [\sin(2\pi f t) + (1/3) \sin(2\pi (3f)t)]$$

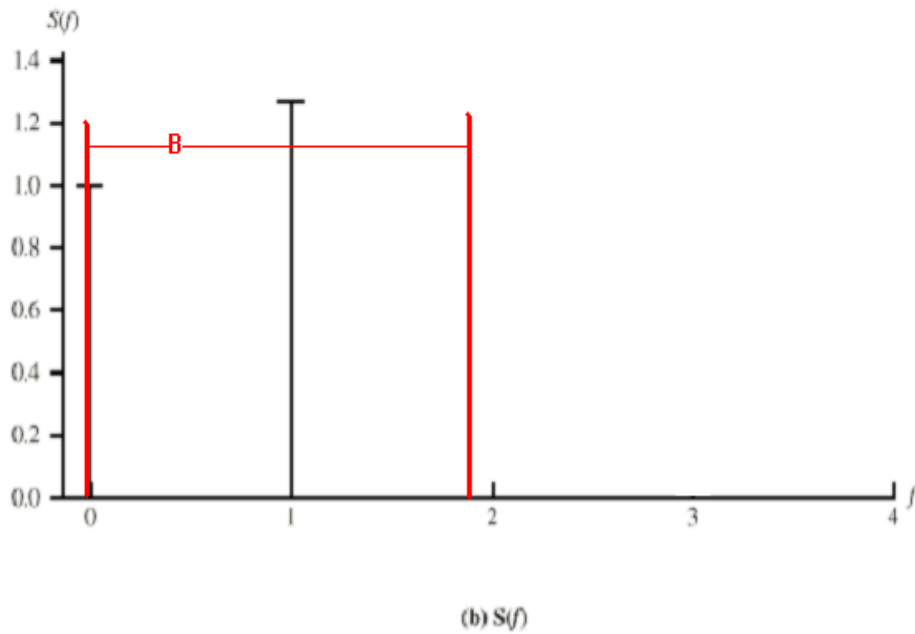
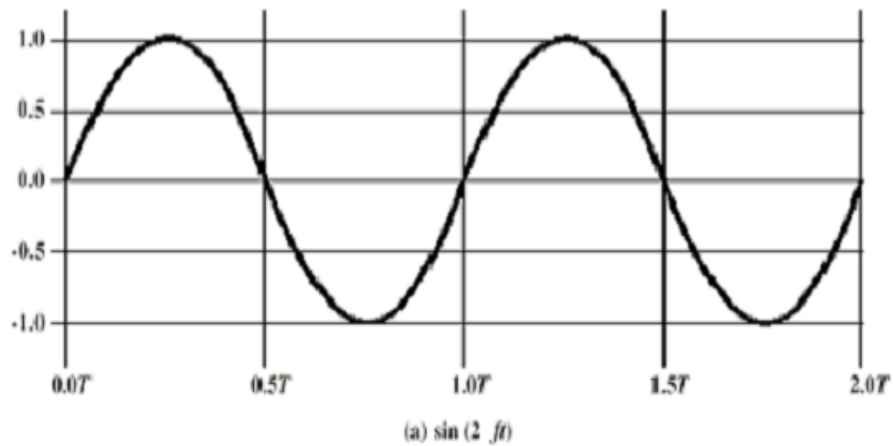


(b) $S(f)$

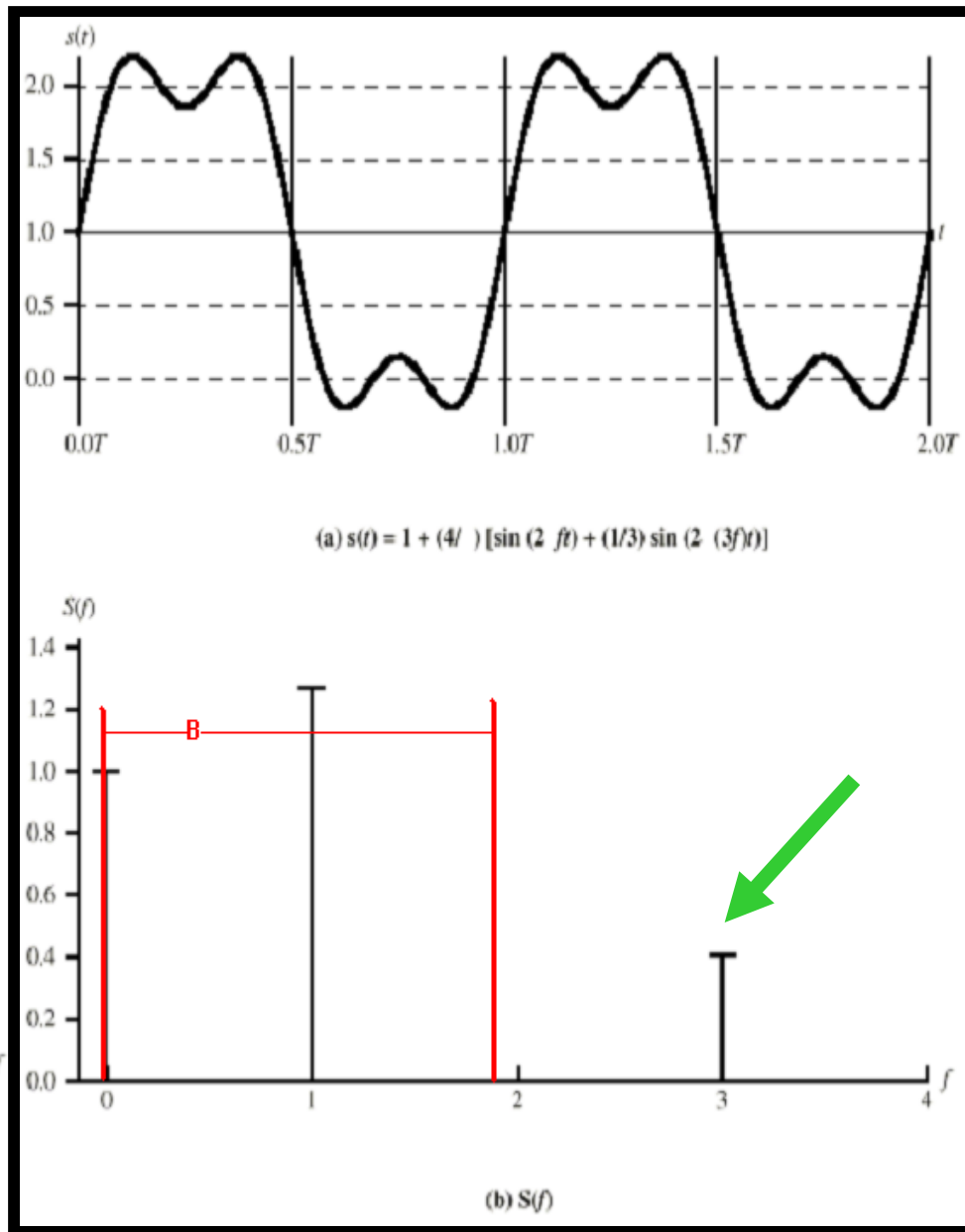


(b) $S(f)$





[Espectro y señal en destino](#)



[Espectro y señal en origen](#)